A7 — Multi-Agent RL: task, goal, and how it was solved

Why every convergence guarantee from A2 onward assumed something that stops being true the moment a second learner enters the environment. The lab notebook is `experiments/a7_multi_agent/README.md`.

> **At a glance**
>
> - Non-stationarity, measured: checkpoints score worse against later opponents than against the one they trained with (mean drop 0.465).
> - Self-play wins (optimal-move rate 0.936 against 0.77 and 0.75), and mostly because a shared table sees twice the states.
> - Independent learners specialise by seat; self-play cannot fail that way.

---

1. The task

`ladder.md` states two items:

1. Two independent Q-learners in a competitive game; observe non-stationarity from each agent's view.
2. Self-play (one agent playing both sides); compare policy quality and stability to independent learners.

2. The goal: what exactly breaks

A2 proved Q-learning converges to Q^* given Robbins–Monro step sizes, infinite visitation, and — silently — a **fixed MDP**. The stochastic approximation argument needs the operator being iterated to stand still.

With two learners it does not. From agent 1's perspective the transition kernel is

$$P_1(s' \mid s, a) = \sum_{a_2} \pi_2(a_2 \mid s)\, P(s' \mid s, a, a_2)$$

which contains π_2. As agent 2 learns, P_1 changes, so agent 1 is solving a **different MDP at every moment**. There is no fixed point for it to converge to, and no contraction argument to make.

Concretely, the operator each agent iterates is

$$(\mathcal{T}_1 Q)(s, a) = \mathbb{E}_{a_2 \sim \pi_2}\left[r(s, a, a_2) + \gamma \sum_{s'} P(s' \mid s, a, a_2) \max_{a'} Q(s', a') \right]$$

which is still a γ-contraction *for fixed* π_2 — so each agent would converge if the other ever stopped. The failure is not that the operator is bad, it is that $\mathcal{T}_1$ is re-indexed by π_2 at every step, and a sequence of contractions with moving fixed points has no reason to converge to any of them.

Worse, the pair need not be heading anywhere at all. The natural solution concept becomes a Nash equilibrium rather than an optimum, and independent Q-learning has no mechanism aiming at one: each agent computes a best response to the other's *current* policy, and best-response dynamics can cycle indefinitely. Convergence of the pair is not merely unproven, it is not generally expected.

Why this phase uses a solvable game. In a two-player game the obvious metric — win rate against the other agent — is nearly uninformative: it says the two are mismatched, not whether either is good. Two agents can trade wins forever while both play badly. Tic-Tac-Toe is small enough to solve exactly (4520 reachable non-terminal positions), which allows an absolute measure:

$$\text{exploitability}(\pi) = \max_{\pi'} V^{\pi',\pi} \; - \; V^{*}$$

the value a *perfect* opponent extracts. Zero means unexploitable. This is the standard measure in game-solving work for exactly this reason, and it is computable here rather than estimated.

3. Self-play, and what it does and does not fix

Self-play uses one policy for both seats. Two things follow immediately:

- **The asymmetry disappears.** There is no possibility of one seat outrunning the other, because there is only one thing to improve.
- **The opponent's strength tracks yours automatically**, which is the curriculum argument usually given for it.

Neither of these makes the problem stationary. The shared policy still bootstraps from values produced by an earlier version of itself, so the target still moves — self-play changes *who* the moving target is, not that there is one. It is a symmetrisation, not a solution, and the honest test is whether the result is close to equilibrium, which exploitability answers directly.

There is also a third effect, rarely stated and — as it turns out here — large: **a shared table receives updates from both seats**, so at equal episode counts self-play collects roughly twice the data per parameter as two independent tables. Any comparison that does not separate this from the curriculum effect is attributing to strategy what is partly just sample count.

4. What was done

Tabular Q-learning on Tic-Tac-Toe, three arms — independent, self-play, and self-play with random starting positions. The third exists as a **control for coverage**: it improves the state distribution by a mechanism that has nothing to do with opponent quality, so if quality tracks it, the curriculum story is incomplete.

Measurement decisions:

- **Exploitability is exact**, by dynamic programming over the game tree, and validated on known cases (a perfect policy scores 0.000; uniform random $+0.925$ first, $+0.995$ second).
- **Graded metrics alongside it** — optimal-move rate over all 4520 positions, and regret against perfect play from random starts — because exploitability is all-or-nothing on a deterministic policy and saturates.
- **Regret rather than raw value** from random starts, since some starting positions are simply better than others and the raw number would measure the start distribution.
- **Non-stationarity as a cross-checkpoint matrix**: freeze seat-1 policies from every checkpoint, play them against seat-2 policies from every checkpoint, compute the value exactly. If each agent's environment were stationary, a row would be flat.

5. Results

Tables in the phase README.

Non-stationarity, measured. Every seat-1 checkpoint scores better against its contemporary opponent than against later ones — mean drop $+0.465$, positive in 6 of 7 rows. The 18k policy goes from $+0.80$ against the opponent it trained alongside to -0.10 against later ones, without changing.

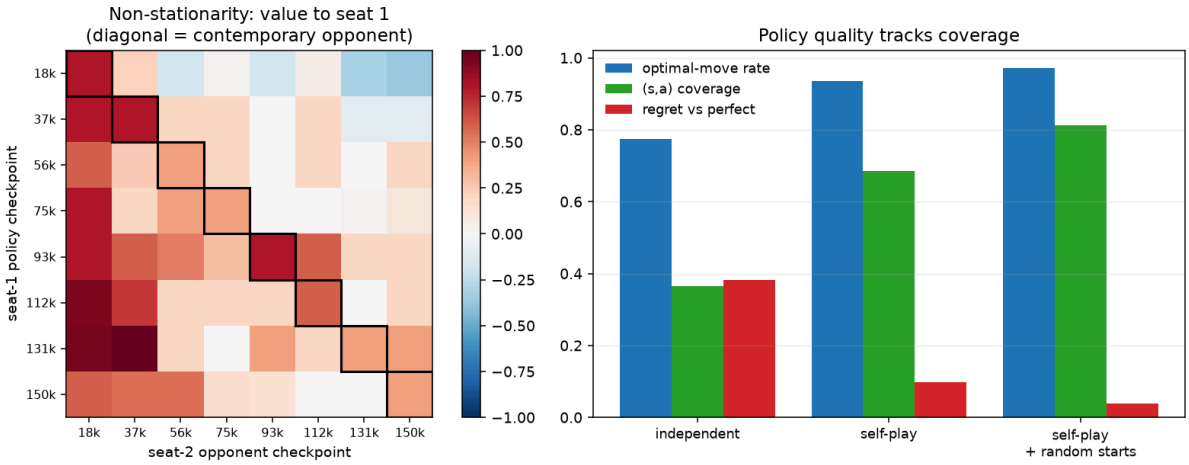


Figure — Left: seat-1 checkpoints scored against every seat-2 checkpoint; the outlined diagonal is each one's contemporary opponent. Right: policy quality tracks state-action coverage across the three training arms.

The row-wise form of the comparison is doing real work. The raw diagonal ($+0.575$) versus off-diagonal ($+0.254$) is confounded: column means fall from $+0.79$ to $+0.13$ across training, so later opponents are genuinely stronger. Comparing each policy to *its own* future opponents controls for that, and the effect survives — which is the difference between "opponents got better" and "this policy was tuned to a specific opponent".

Self-play wins, substantially for a reason that is not the usual story. Optimal-move rate 0.936 against $0.773/0.749$; regret 0.099 against 0.383. But state-action coverage is 0.684 against 0.320–0.364 — almost exactly double, which is what a shared table updated from both seats should produce. The random-starts arm confirms the direction: it raises coverage to 0.813 by a mechanism unrelated to opponent quality, and quality follows (0.972, regret 0.038). Across all three arms, quality tracks coverage.

So self-play's advantage here decomposes into at least three parts — twice the updates, a symmetric curriculum, and no seat specialisation — and the experiment separates the first from the others rather than reporting a single number.

Independent learners are seat-specialists. The seat-2 table, made to play first, scores $+0.92$ exploitability against $+0.00$ for the seat-1 table. Each learned one side of the game only. Self-play cannot have this failure at all, which is a genuine advantage independent of data volume.

And nothing reaches equilibrium. Second-player exploitability is $+1.00$ for every arm, including the best one at 97.2% optimal moves. The reason is worth stating precisely because it is not the obvious one: it is not a coverage failure — exploitability stayed pinned at 1.00 while coverage rose $0.509 \to 0.947$ in a separate sweep. Walking the exploiter's best line to the first mistake shows that at the critical position the never-tried actions sit at their zero initialisation while every tried action is *negative*, because those moves lose. **Zero-initialised Q is optimistic in a zero-sum game.** Every true value satisfies $Q^*(s,a) \in [-1,+1]$ with the great majority at ≤ 0, so for an untried action

$$Q_0(s,a) = 0 \;\geq\; Q^*(s,a) \quad \text{for most } (s,a)$$

and the greedy policy is systematically drawn toward whatever it has not tried — exactly where a best-response opponent steers. This is A3's optimism with the sign flipped into a different failure: there an over-valued action was merely pursued, here it is *selected against by an adversary*. Exploitability demands correctness along the whole line, so one such state is fatal.

That is the sharpest statement of this phase's lesson: a policy that is optimal in 97.2% of positions loses *every* game to an opponent allowed to choose the line. Average-case competence and worst-case exploitability are different quantities, and in adversarial settings only the second one is the score.

6. What this sets up

- **B4's reward hacking** is the same structure with the roles renamed: a policy optimised against a proxy, then evaluated by something that searches for where the proxy is wrong. The exploiter here is a best response; there it is the optimiser itself finding the reward model's blind spots.
- **A8's inverse RL** shares the identifiability problem — many reward functions explain the same behaviour, just as many policies share the same win rate against a fixed opponent while differing wildly in exploitability.

7. Where these items come from

item	primary source	why it was picked
1 — independent Q-learners	Tan 1993, the standard reference for independent learners in a multi-agent setting	It is the minimal way to break A2's stationarity assumption — change nothing about the algorithm, add a second learner — so the failure is attributable to the setting rather than to any design choice.
2 — self-play	OpenAI's competitive self-play work; MADDPG is the ladder's reading for the centralised-critic alternative	Self-play is the standard response, and the interesting question is not whether it helps but *why* — which requires separating curriculum from sample count, and asking whether it reaches equilibrium at all.

The **lab** pairing is the ladder's suggested PettingZoo tutorials. They are not used: an exactly solvable game buys absolute measurement, which in a two-player setting is worth more than environment variety, and the reasoning is recorded in the phase README.

The **goal statement** in section 2 is quoted from `ladder.md`. Section 3's decomposition of self-play's advantage into three separable effects, and the random-starts control that tests it, are this repo's rather than claims from the sources.